Akoizo Original Practice - 2027 Rotation Form 3

Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Two second-order streams create…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Second order

B. Fourth order

C. First order

D. Third order

Answer: _____

2. (2 points) Hydrograph area represents…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Only peak stage

B. Only velocity

C. Only basin area

D. Volume

Answer: _____

3. (2 points) Impervious connectivity can affect…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only mineral hardness

B. Only gene frequency

C. No water process

D. Runoff timing

Answer: _____

4. (2 points) Groundwater divides always match surface divides?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Always

B. Only on maps

C. Only in winter

D. No

Answer: _____

5. (2 points) A 100-year label means exact spacing?

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Always

B. Only for rivers

C. Only after one flood

D. No

Answer: _____

6. (2 points) Load rate needs…

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Only pH

B. Only map area

C. Only color

D. Concentration and discharge

Answer: _____

7. (2 points) Radial drainage can suggest…

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

A. Only flat sediment

B. Only a straight fault

C. No topography

D. An elevated central feature

Answer: _____

8. (2 points) Stage and discharge are…

Unit: Streamflow, hydrographs, and floods.
Worked-case data: A triangular hydrograph has base 4 h and peak 10 m^3/s above zero. Find volume.

A. Identical units

B. Always equal numbers

C. Unrelated by any model

D. Related but different quantities

Answer: _____

9. (2 points) Before/after change proves causality alone?

Unit: Land use, water pollution, and hydrological impacts.
Worked-case data: Concentration is 2 mg/L and discharge is 3 m^3/s. Find load rate.

A. Always

B. Only after one storm

C. Only in a reservoir

D. No

Answer: _____

10. (4 points) Why specify the outlet?

Unit: Watersheds, stream order, and drainage patterns.
Worked-case data: A first-order stream joins a third-order stream, then another third-order joins. Find successive orders.

Answer: _____
_____
_____

11. (2 points) A freshwater catchment receives 130 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

Answer: _____
_____
_____

12. (2 points) A rectangular stream is 7 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

Answer: _____
_____
_____

13. (2 points) A river channel is 15 km long between points 11 km apart along the valley. Find sinuosity.

Answer: _____
_____
_____

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

Answer: _____
_____
_____

15. (2 points) Use Q=K A i with K=0.001 m/s, A=23 m^2 and i=0.02. Find saturated Darcy discharge.

Answer: _____
_____
_____

16. (2 points) A lake stores 3000 m^3 and exports 300 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

Answer: _____
_____
_____

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

Answer: _____
_____
_____

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

Answer: _____
_____
_____

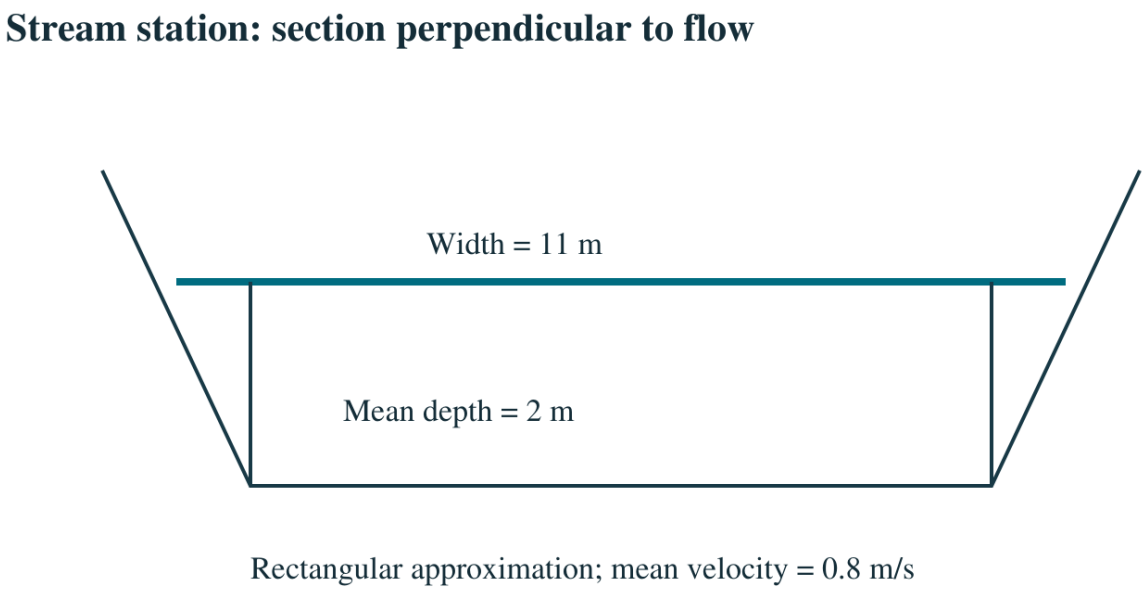
19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

Answer: _____
_____
_____

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Estimate wetted cross-sectional area using the indicated rectangular approximation.



F1. Stream cross section and measured velocity (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate stream discharge from this section.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

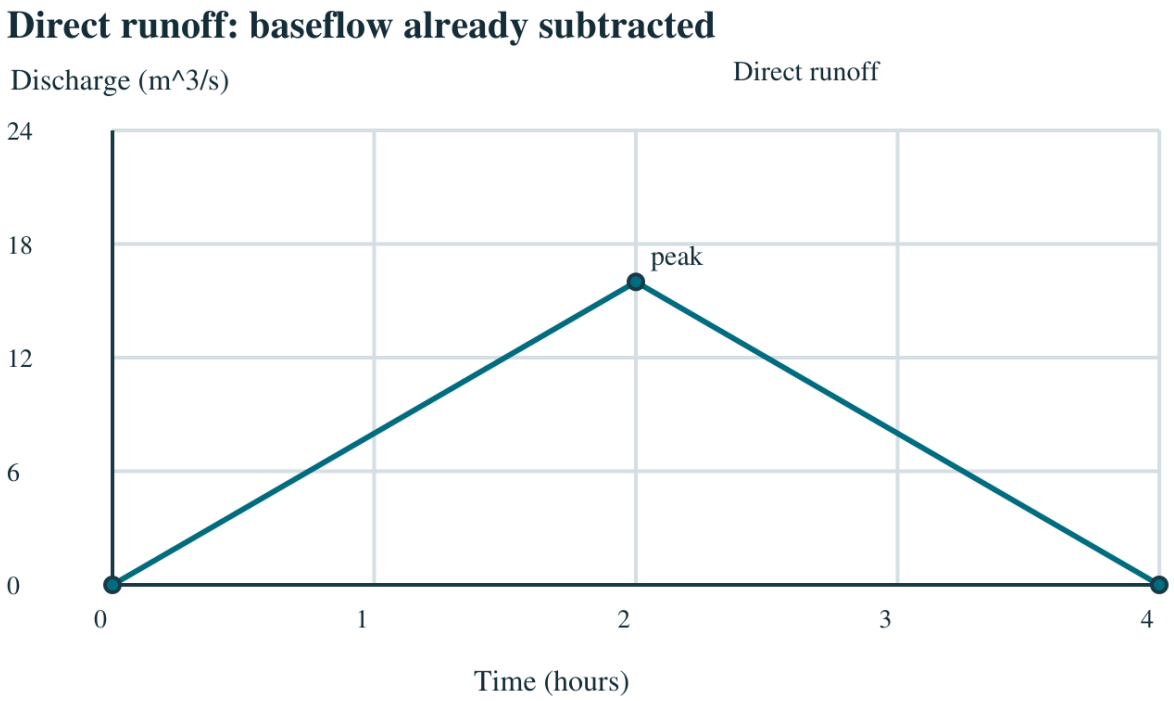
Answer: _____
_____
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23. (2 points) Use Figure F1. The water rises while width and mean velocity remain fixed. Explain the predicted discharge change.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Read the time of peak direct runoff.



F2. Storm hydrograph (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Find the event runoff volume. Convert hours to seconds.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

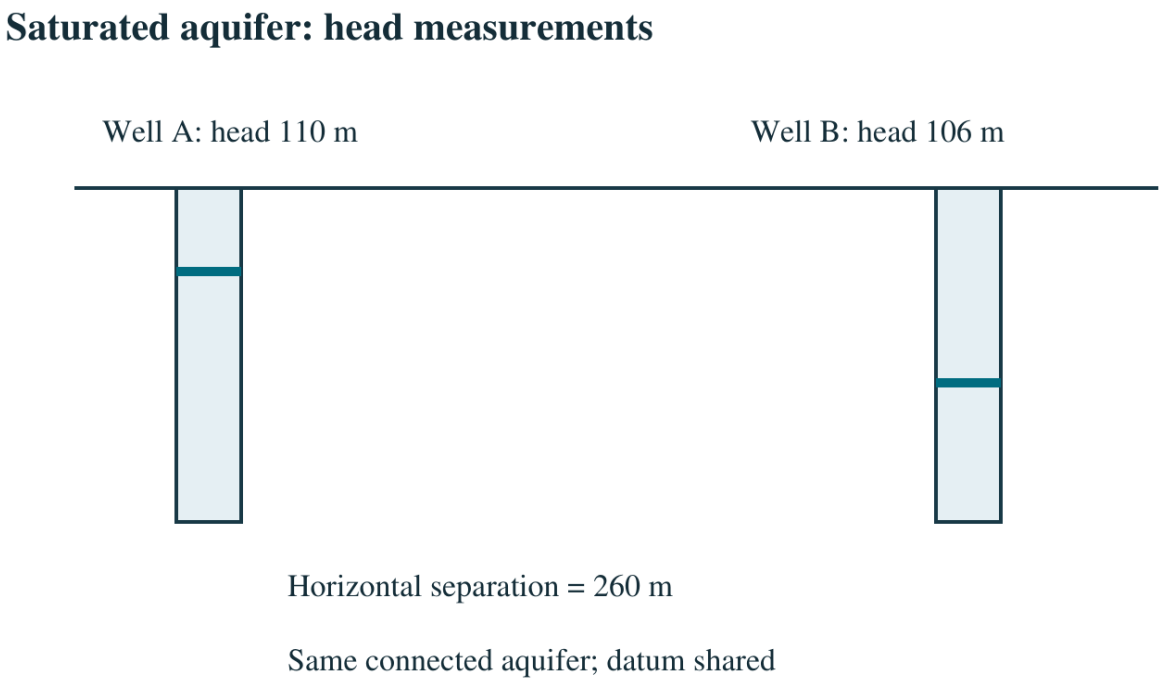
Answer: _____
_____
_____

26. (2 points) Use Figure F2. A wetland spreads the same runoff volume over more time. Describe how its hydrograph could differ.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate hydraulic-gradient magnitude.



F3. Groundwater heads in two wells (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Predict the horizontal flow direction and identify the controlling measurement.

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

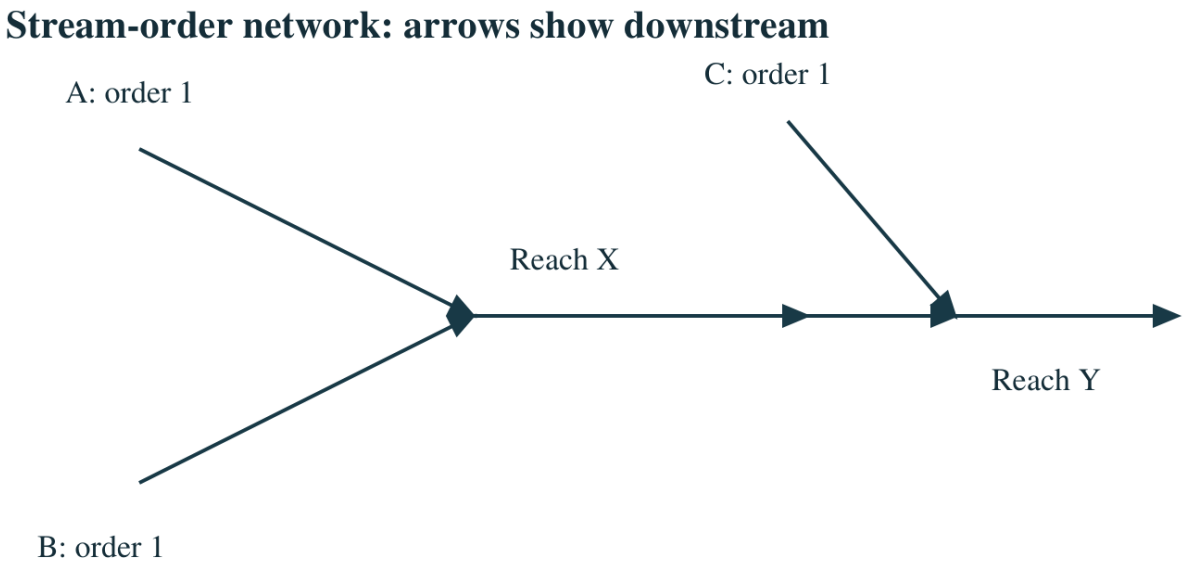
Answer: _____
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29. (2 points) Use Figure F3. Why cannot the diagram alone determine groundwater discharge?

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Give the Strahler order of reach X.



F4. Tributary network (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Give the Strahler order of reach Y.

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. If C were second order instead, how would Y change?

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____